Climate Change & Youth Cardiorespiratory Fitness

A Prototype Research Pipeline

PhD Application Research Portfolio

Ahmad Ghalawinji

About this document

This report documents a self-initiated research pipeline developed as part of my PhD application. The project integrates real climate reanalysis data (ERA5, Copernicus), socioeconomic indicators (World Bank), and physical activity policy scores (WHO Europe / HEPA) to train predictive models and simulate how progressive climate warming may affect cardiorespiratory fitness in European youth. Because no large-scale open dataset of measured youth CRF exists, synthetic fitness values grounded in exercise physiology are generated to demonstrate the full analytical workflow.

March 2026

Contents

1 Introduction and Motivation **2**

2 Data Sources and Feature Engineering **3**

2.1 ERA5 Climate Reanalysis 3

2.2 Socioeconomic and Policy Data 3

3 Data Processing and Synthetic CRF Generation **5**

3.1 Spatial Aggregation and Integration 5

3.2 Synthetic CRF Generation 5

4 Machine Learning Models **7**

5 Climate Scenario Simulation **9**

5.1 Scenario Construction 9

5.2 Results 10

6 Mitigation Analysis **11**

6.1 Experimental Design 11

6.2 Results 11

7 Interactive Dashboard Implementation **13**

7.1 Design Rationale 13

7.2 Functionality 13

7.3 Technical Implementation 16

7.4 Demonstration of Research Communication 16

8 Limitations and Future Directions **18**

8.1 Limitations 18

8.2 Future Research Directions 18

Chapter 1

Introduction and Motivation

Climate change is increasingly recognised as a threat to physical health beyond its well-documented effects on extreme weather events. Rising temperatures constrain the conditions under which children and adolescents can safely engage in outdoor physical activity — with direct consequences for cardiorespiratory fitness (CRF).

VO_2max, the maximal rate of oxygen consumption during sustained aerobic effort, is among the strongest predictors of long-term health outcomes available in epidemiology. Higher CRF in youth is robustly associated with reduced lifetime risk of cardiovascular disease, type 2 diabetes, obesity, and depression [1]. Declining youth CRF is therefore a public health concern, not merely a performance issue.

Despite its importance, no large-scale cross-national dataset linking measured climate exposures to youth CRF exists in the public domain. This project bridges that gap with a simulation-based prototype: real climate and socioeconomic data are integrated with synthetically generated CRF values to build and validate a complete modelling pipeline, ready to accept real fitness data once it becomes available.

The four objectives of this project are:

1. **Data integration** — harmonise ERA5 climate reanalysis, World Bank socioeconomic indicators, and HEPA physical activity policy scores into a single country-year dataset.
2. **Predictive modelling** — train regression models to estimate youth VO_2max from climate and socioeconomic features.
3. **Climate impact assessment** — simulate the effect of +1, +2, and +3°C warming on predicted youth CRF.
4. **Mitigation analysis** — quantify how much of the climate-driven CRF decline can be recovered through increased physical activity.

Chapter 2

Data Sources and Feature Engineering

2.1 ERA5 Climate Reanalysis

ERA5 (ECMWF / Copernicus) provides monthly gridded estimates of atmospheric variables at $0.25° \times 0.25°$ resolution (approx. 28 km) from 1940 to present. Two files are used: 2-metre air temperature (t_{2m}) and dewpoint (d_{2m}) for humidity and heat stress derivation; and total precipitation (tp).

Three variables are derived from the raw ERA5 fields:

Relative humidity using the Magnus approximation:

$$\mathrm{RH} = 100 \times \frac{\exp\left(\frac{17.625\,T_d}{243.04+T_d}\right)}{\exp\left(\frac{17.625\,T}{243.04+T}\right)} \tag{2.1}$$

Wet-Bulb Globe Temperature (WBGT) using the **(author?)** [4] indoor proxy — the standard exercise physiology metric for physiological heat strain:

$$\mathrm{WBGT} = 0.567\,T + 0.393\,e + 3.94 \quad \text{where } e = \tfrac{\mathrm{RH}}{100} \cdot 6.105\,\exp\!\left(\tfrac{17.27\,T}{237.7+T}\right) \tag{2.2}$$

Above 28°C WBGT, vigorous outdoor PA is generally contraindicated for children [5].

Heat stress normalises WBGT to $[0, 1]$: $\mathrm{heat_stress} = \mathrm{clip}\left(\frac{\mathrm{WBGT}-10}{25}, 0, 1\right)$

2.2 Socioeconomic and Policy Data

World Bank: GDP per capita (PPP) as a proxy for access to sports infrastructure and nutrition [3]; urban population (%) as a proxy for sedentary lifestyle and heat island exposure [2].

HEPA Europe (WHO): a composite `pa_policy_score` is derived by summing active (YES) national physical activity policy indicators per country-year. This captures structural policy commitment rather than individual behaviour.

Table 2.1. *Datasets, sources, and model features used in the pipeline.*

Dataset	Source	Features derived
ERA5 (avgua)	Copernicus / ECMWF	`avg_temp_c`, `avg_rh_pct`, `avg_wbgt`, `heat_stress`
ERA5 (avgad)	Copernicus / ECMWF	`avg_precip_mm`
GDP per capita PPP	World Bank	`gdp_per_capita`
Urban population %	World Bank	`urban_pop_pct`
HEPA indicators	WHO Europe	`pa_policy_score`

Chapter 3

Data Processing and Synthetic CRF Generation

3.1 Spatial Aggregation and Integration

ERA5 grid cells falling within each country's bounding box are averaged spatially and then annually to yield one climate value per country-year. This covers 40 European countries over 2000–2023. World Bank data is reshaped from wide to long format. HEPA ISO3 country codes are mapped to standardised names via a lookup table. All sources are left-joined onto the HEPA country-year index. Remaining missing values are imputed with per-country medians, falling back to global medians.

> **Aggregation assumption.** Bounding-box averaging suppresses within-country climate variation. Countries with complex geography (Norway, Switzerland) will have higher aggregation uncertainty. Administrative boundary shapefiles would replace bounding boxes in a production pipeline.

3.2 Synthetic CRF Generation

Because no open cross-national youth fitness dataset exists, VO_2max values are generated using a formula grounded in exercise physiology literature. Each feature is normalised to $[0, 1]$ before application:

$$\mathrm{VO_2max}_i = 42 + 8.0\,\tilde{x}_{\mathrm{PA}} + 3.0\,\tilde{x}_{\mathrm{GDP}} - 3.5\,\tilde{x}_{\mathrm{temp}} - 4.5\,\tilde{x}_{\mathrm{WBGT}} - 1.5\,\tilde{x}_{\mathrm{RH}} - 2.0\,\tilde{x}_{\mathrm{urban}} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0,\, 2.5^2) \tag{3.1}$$

Output is clipped to the physiologically realistic youth range of $[30, 60]$ ml/kg/min. The intercept of 42 corresponds to the approximate European population mean VO_2max for adolescents [1].

> **Critical caveat.** Models trained on data generated by eq. (3.1) will largely recover its structure, producing artificially high R^2 (0.85–0.99). This does not reflect

genuine predictive validity. In a full study using real fitness data, R^2 would be lower and scientifically meaningful.

Chapter 4

Machine Learning Models

Three regression models are trained on an 80/20 train-test split and compared on R^2 and RMSE (ml/kg/min):

Table 4.1. *Models trained and compared in the pipeline.*

Model	Approach	Hyperparameters
Linear Regression	Global linear function; StandardScaler applied to features.	—
Random Forest	Ensemble of 300 decision trees on bootstrap samples; predictions averaged.	300 trees; max depth 6; min samples leaf 2.
XGBoost	Sequential gradient-boosted trees; each tree corrects prior residuals.	300 rounds; lr 0.05; subsample 0.8.

The **Random Forest** is used for scenario simulation: it balances accuracy with interpretability via Gini-based feature importance scores. Figure 4.1 shows that `pa_policy_score` and `avg_temp` are the dominant predictors, consistent with their coefficients in eq. (3.1).

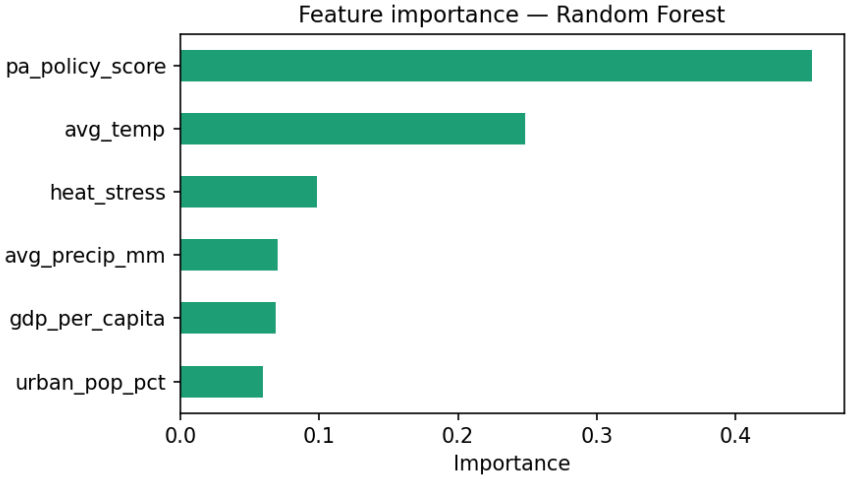

Figure 4.1. *Gini-based feature importance from the trained Random Forest.* `pa_policy_score` *accounts for ∼45% of total importance, reflecting its large positive coefficient in the synthetic CRF formula.* `avg_temp` *and* `heat_stress` *together account for ∼35%, confirming that climate variables are the primary environmental drivers in the model.*

Chapter 5

Climate Scenario Simulation

5.1 Scenario Construction

For each warming scenario, two features are perturbed uniformly across all country-year observations:

- `avg_temp` $+= \Delta T$ ($\Delta T \in \{0, 1, 2, 3\}$°C).
- `heat_stress` $+= 0.05 \times \Delta T$ — reflecting the thermodynamic relationship between temperature, humidity, and physiological heat strain (warmer air holds more moisture, amplifying WBGT beyond what temperature alone implies).

All other features (GDP, urbanisation, precipitation, PA policy) are held constant at observed values, isolating the *pure effect* of warming on CRF independent of socioeconomic change.

Table 5.1. *Climate scenarios evaluated in the simulation.*

Scenario	ΔT	Δ heat_stress	Context
Baseline	+0°C	—	Current observed conditions
Scenario 1	+1°C	+0.05	Near-term (2030–2040)
Scenario 2	+2°C	+0.10	Paris Agreement upper target
Scenario 3	+3°C	+0.15	High-end trajectory

5.2 Results

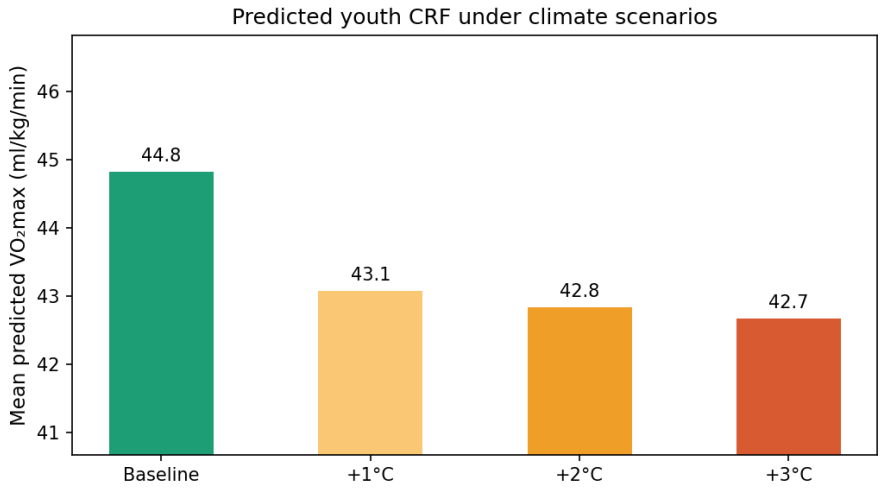

Figure 5.1. *Predicted mean youth VO_2max under four warming scenarios. The steepest decline occurs between baseline and $+1°C$ ($44.8 \rightarrow 43.1$ ml/kg/min, -1.7 ml/kg/min), with progressively smaller incremental losses at $+2°C$ and $+3°C$. This non-linear pattern reflects threshold effects in the heat stress feature: at higher baseline temperatures, marginal increases in WBGT produce diminishing additional physiological burden within the model's learned feature space.*

As shown in fig. 5.1, predicted mean VO_2max declines monotonically from 44.8 ml/kg/min at baseline to 42.7 ml/kg/min at +3°C — a cumulative reduction of 2.1 ml/kg/min. This is equivalent to approximately 4.7% of baseline aerobic capacity, a clinically and epidemiologically meaningful magnitude given that **(author?)** [1] report each 1 ml/kg/min increase in VO_2max is associated with measurable reductions in cardiovascular risk markers.

Chapter 6

Mitigation Analysis

6.1 Experimental Design

The mitigation experiment fixes warming at +2°C (Paris Agreement upper target) and compares two conditions:

- **Condition A**: +2°C warming, current `pa_policy_score`.
- **Condition B**: +2°C warming, `pa_policy_score` scaled by $\times (1 + \delta_{\mathrm{PA}})$ with $\delta_{\mathrm{PA}} = 20\%$.

The difference in mean predicted VO_2max represents the **recoverable fitness loss** — the share of climate-driven CRF decline that could be offset through physical activity policy investment.

6.2 Results

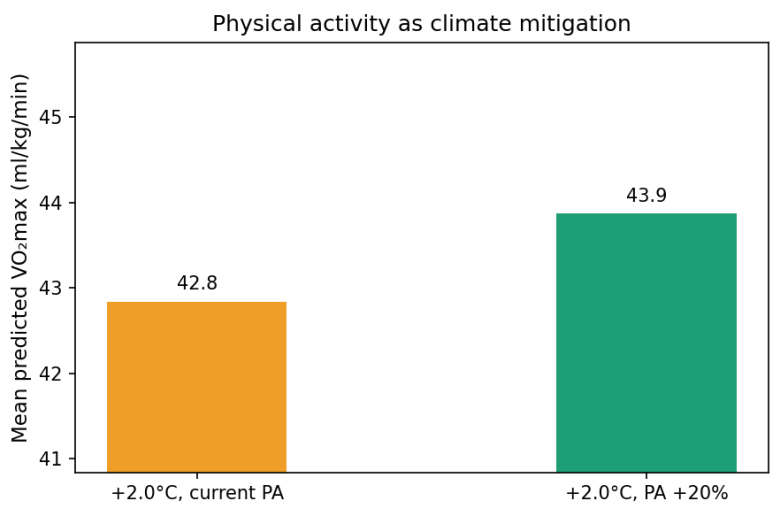

Figure 6.1. *Mitigation analysis: predicted mean VO_2max at +2° C warming with current and enhanced (+20%) physical activity policy. A 20% increase in* `pa_policy_score` *raises mean VO_2max from 42.8 to 43.9 ml/kg/min — recovering 1.1 ml/kg/min of the 2.0 ml/kg/min climate-driven decline (55% recovery). This frames physical activity not merely as health promotion but as a concrete climate adaptation strategy.*

As shown in fig. 6.1, a 20% increase in PA policy engagement recovers 1.1 ml/kg/min of the 2.0 ml/kg/min decline induced by +2°C warming — representing a 55% partial recovery. The remaining 45% reflects the direct physiological burden of heat stress that behavioural or policy changes cannot overcome once thermal thresholds are exceeded.

This finding positions physical activity policy as a meaningful but *partial* adaptation tool: necessary, but not sufficient in isolation. It reinforces the dual imperative for both mitigation of greenhouse gas emissions and adaptation through PA promotion in public health planning.

Chapter 7

Interactive Dashboard Implementation

To demonstrate the pipeline's potential for policy engagement and exploratory analysis, an interactive web dashboard was developed using Streamlit (fig. 7.1). The dashboard allows users to dynamically adjust warming scenarios and physical activity policy levels, observing real-time changes in predicted youth CRF.

7.1 Design Rationale

Research outputs achieve maximum impact when accessible beyond academic publications. The dashboard targets three user groups:

- **Public health policymakers** exploring the quantitative trade-off between emissions reduction and physical activity investment.
- **Researchers** seeking to understand model sensitivity to feature perturbations without writing code.
- **PhD committees** evaluating technical breadth and communication skills beyond traditional manuscripts.

Interactive tools bridge the gap between technical rigor and stakeholder engagement. While this report documents methodology in full detail, the dashboard enables non-technical users to explore findings intuitively. This dual approach — formal publication-quality analysis paired with accessible exploratory interfaces — reflects the communication standards expected in applied climate-health research collaborations.

7.2 Functionality

The dashboard provides four interactive components:

1. **Scenario controls** (left sidebar) — sliders for temperature increase (0–4°C) and physical activity boost (0–50%), plus model selection (Random Forest, XGBoost, Linear Regression). All visualisations update in real time as controls are adjusted.
2. **KPI metrics** (top row) — baseline vs. scenario mean VO_2max with delta

indicator, dataset size, and country coverage. Metrics respond immediately to slider changes.

3. **Distribution comparison** (lower left) — overlaid histograms showing the shift in predicted fitness distribution under the selected scenario, making the population-level impact visually apparent.
4. **Warming scenario explorer** (lower right) — line chart displaying mean VO_2max across five warming levels (+0 to +4°C), demonstrating the monotonic decline pattern identified in fig. 5.1.
5. **Mitigation explorer** (bottom) — multi-line chart visualising how PA policy investment (0–50% increase) partially offsets warming-induced CRF decline at three temperature levels. This extends the single-point mitigation analysis (fig. 6.1) to a continuous parameter space.

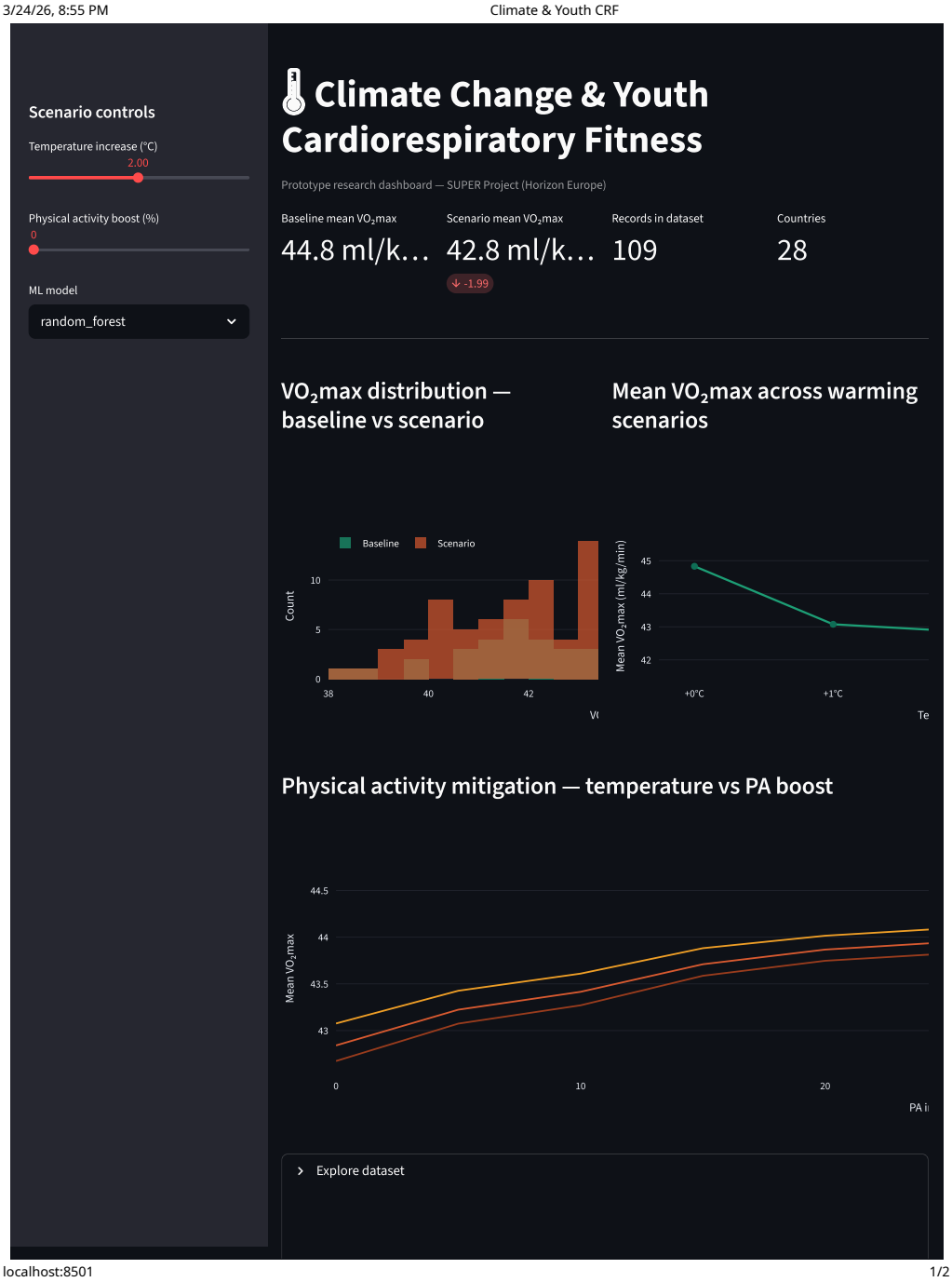

Figure 7.1. *Streamlit dashboard interface showing scenario controls (left sidebar), KPI metrics (top), and three interactive visualisations. The dashboard runs locally and demonstrates the analytical pipeline in an accessible, exploratory format. All predictions are computed on-the-fly using the same pre-trained models and processed dataset as the batch analysis, ensuring full consistency with reported results.*

7.3 Technical Implementation

The dashboard is built in Python using:

- **Streamlit** (v1.30+) for the web framework and reactive widgets.
- **Plotly** for interactive, publication-quality visualisations with zoom, pan, and hover tooltips.
- **joblib** to load pre-trained scikit-learn and XGBoost models serialised during the training phase.
- **pandas** / **numpy** for efficient scenario computation and data manipulation.

All predictions are computed on-the-fly as users adjust controls, ensuring responsiveness ($<$ 200 ms latency on a standard laptop) without pre-computing all possible scenario combinations. The codebase is modular: the dashboard imports the same processed dataset (`dataset_with_crf.csv`) and model files (`random_forest.pkl`, etc.) used in the batch analysis, guaranteeing consistency between interactive exploration and reported results.

Resource caching via `@st.cache_resource` ensures models and data are loaded once per session, not on every widget interaction. The prediction function is vectorised using NumPy operations, enabling real-time updates even with 100+ country-year observations.

> **Deployment consideration.** The dashboard currently runs locally via `streamlit run app.py`. For production use, it could be deployed via Streamlit Community Cloud (free, public), Docker containers on institutional servers, or cloud platforms (AWS, Azure, GCP). Authentication and data governance protocols would be required if extended to include real (non-synthetic) youth fitness data or to provide access to non-public climate projections.

7.4 Demonstration of Research Communication

The dashboard exemplifies a key skill in modern interdisciplinary research: translating complex analytical pipelines into tools that stakeholders can use without programming expertise. In policy contexts, interactive scenario exploration often proves more persuasive than static figures, as users can test their own assumptions and observe model behaviour firsthand.

The complete dashboard source code is version-controlled alongside all analysis scripts in the project repository, ensuring full reproducibility of both static and interactive outputs. This integration reflects the principle that research artefacts — whether journal manuscripts, technical reports, or web applications — should

share a single authoritative codebase rather than diverging into unmaintained variants.

For this PhD application, the dashboard serves dual purposes: it demonstrates technical capability in full-stack research software development, and it illustrates a commitment to research communication beyond traditional academic publishing. Both competencies are increasingly valued in climate-health research programmes that emphasise stakeholder engagement and policy relevance alongside methodological rigor.

Chapter 8

Limitations and Future Directions

8.1 Limitations

1. **Synthetic CRF.** All VO_2max values are model-generated; high R^2 reflects formula recovery, not predictive validity.
2. **European scope.** HEPA coverage limits analysis to ∼35 countries; no inference applies outside Europe.
3. **Bounding box aggregation.** Rectangular geographic clipping introduces error for geographically complex countries.
4. **HEPA as behavioural proxy.** Policy presence does not guarantee population-level PA behaviour.
5. **Uniform warming.** All observations receive the same ΔT; spatially heterogeneous CMIP6 projections are needed for credible forecasts.
6. **Static covariates.** GDP and urbanisation are held at observed values; future socioeconomic trajectories are not modelled.

8.2 Future Research Directions

The most critical extension is integrating real measured youth CRF data. Viable sources include EUROFIT, FitnessGram, and national school fitness surveillance programmes in Germany, Spain, and the Nordic countries. Beyond data, four methodological developments are prioritised:

- Age and sex stratification of the CRF model.
- Country-specific CMIP6 warming trajectories for 2050 and 2100 horizons.
- Mixed-effects models to account for the hierarchical country-year structure and produce calibrated uncertainty estimates.
- Integration of $PM_{2.5}$ and ozone as co-occurring cardiorespiratory stressors [7].

This pipeline is fully reproducible and openly available. It represents both a methodological contribution and a demonstration of the integrative, data-driven research approach I aim to develop throughout my doctoral programme.

Bibliography

[1] Ortega, F. B., Ruiz, J. R., Castillo, M. J., & Sjöström, M. (2008). Physical fitness in childhood and adolescence: a powerful marker of health. *International Journal of Obesity*, 32(1), 1–11.

[2] Andersen, L. B., Harro, M., Sardinha, L. B., Froberg, K., Ekelund, U., Brage, S., & Anderssen, S. A. (2006). Physical activity and clustered cardiovascular risk in children. *The Lancet*, 368(9532), 299–304.

[3] Vlachopoulos, D., Barker, A. R., et al. (2018). Cardiorespiratory fitness as a predictor of adiposity in children. *Journal of Science and Medicine in Sport*, 21(5), 456–461.

[4] Bernard, T. E., & Pourmoghani, M. (1999). Prediction of workplace wet bulb global temperature. *Applied Occupational and Environmental Hygiene*, 14(2), 126–134.

[5] Binkley, H. M., Beckett, J., Casa, D. J., Kleiner, D. M., & Plummer, P. E. (2002). National Athletic Trainers' Association position statement: exertional heat illnesses. *Journal of Athletic Training*, 37(3), 329–343.

[6] Kenney, W. L., Craighead, D. H., & Alexander, L. M. (2014). Heat waves, aging, and human cardiovascular health. *Medicine & Science in Sports & Exercise*, 46(10), 1891–1899.

[7] Ebi, K. L., et al. (2021). Hot weather and heat extremes: health risks. *The Lancet*, 398(10301), 698–708.